

MED-ASSIST AI

Medical Symptom Analysis & Disease Prediction System

**Infosys SpringBoard 7.0
Project Documentation**

**Done by
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CHAPTER 1 — PROJECT OVERVIEW & OBJECTIVES

1.1 Introduction

Healthcare is one of the most important domains where technology and artificial intelligence can provide meaningful assistance. The increasing availability of digital health data, machine learning algorithms, cloud computing, and web technologies has created opportunities for developing intelligent healthcare support systems.

MedAssist AI: Medical Symptom Analysis & Disease Prediction System is an AI-powered healthcare assistance platform designed to analyze symptoms provided by users and identify possible health conditions. The system combines symptom analysis, disease prediction, risk assessment, healthcare recommendations, patient history management, and health report generation into a centralized platform. The proposed platform can be useful for patients, healthcare providers, clinics, hospitals, telemedicine platforms, and healthcare organizations. The original project specification specifically identifies these groups as potential users of the platform.

1.2 Project Background

Traditional healthcare diagnosis generally requires a patient to visit a healthcare professional, describe symptoms, undergo examination, and possibly complete laboratory or diagnostic tests. While this process is essential for proper diagnosis, users may sometimes be uncertain about the significance of their symptoms or the urgency of seeking medical attention. MedAssist AI follows this concept by accepting symptoms and relevant patient information as input and processing them through an intelligent prediction workflow.

The system can perform the following major activities:

- User registration and authentication
- Patient profile management
- Symptom selection
- Symptom validation
- Symptom correlation analysis
- Disease prediction
- Prediction confidence scoring
- Health risk assessment
- Severity evaluation

- Healthcare recommendations
- Preventive care suggestions
- Patient history management
- Health report generation
- Analytics and visualization

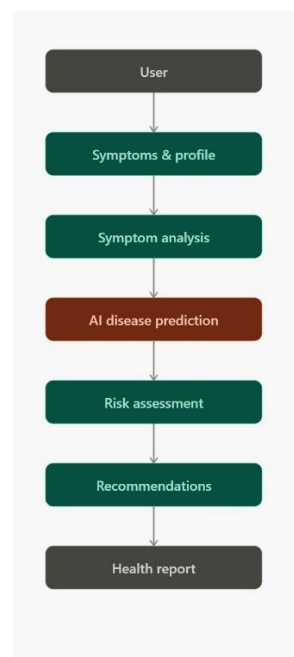
These functions correspond to the modules identified in the project specification.

1.3 Project Vision

The vision of MedAssist AI is to create a centralized intelligent healthcare assistance platform capable of converting user-provided symptoms and health information into understandable preliminary healthcare insights.

The platform aims to make healthcare information more accessible while encouraging users to seek professional medical attention whenever required.

The overall workflow can be represented as:



1.4 Project Objectives

The major objectives of the project are:

Objective 1: Symptom Analysis

The system should allow users to select or enter symptoms and analyze their relationships with potential diseases.

Objective 2: Disease Prediction

The platform should use machine learning classification techniques to identify possible diseases based on symptom combinations.

Objective 3: Risk Assessment

The system should evaluate the potential severity or health risk associated with the user's information.

Objective 4: Healthcare Recommendations

The platform should provide preliminary healthcare recommendations, preventive suggestions, lifestyle guidance, and follow-up guidance.

Objective 5: Health Report Generation

The system should generate understandable reports containing prediction results, risk information, and patient health summaries.

Objective 6: Patient History Management

The platform should maintain relevant patient profiles and historical assessment information.

Objective 7: Analytics

The system should provide analytics dashboards for examining disease predictions, symptom trends, and healthcare-related insights.

Objective 8: Secure Access

Authentication and role-based access control should ensure that different categories of users access appropriate platform features.

1.5 Expected Outcomes

The expected outcomes of MedAssist AI include:

1. An AI-powered medical symptom analysis platform.
2. A functional disease prediction workflow.
3. A risk assessment mechanism.
4. A recommendation engine.
5. Patient history management.
6. Health report generation.

7. Analytics dashboards.
8. Authentication and role-based access.
9. Containerized deployment using Docker.
10. Cloud deployment using platforms such as AWS or Azure.

These expected outcomes are directly aligned with the project specification.

1.6 Advantages of the Proposed System

Accessibility- Users can obtain preliminary health insights through a digital platform.

Early Awareness- The system can help users recognize possible health conditions at an early stage.

Centralized Healthcare Information - Patient profiles, symptom assessments, predictions, risk assessments, and reports can be managed within a single platform.

AI-Assisted Analysis - Machine learning can identify relationships between symptoms and disease categories.

Risk Identification - The system can categorize potential health risks and identify cases requiring higher attention.

Data-Driven Healthcare - The system uses structured healthcare datasets for model development and analysis.

CHAPTER 2 — PROBLEM STATEMENT & SCOPE

2.1 Problem Statement

People frequently experience symptoms without knowing whether those symptoms indicate a minor condition or a potentially serious health problem. Searching for symptoms through unrelated online sources can produce confusing or inconsistent information.

Healthcare organizations also handle large amounts of patient information, making systematic analysis and monitoring increasingly important.

Therefore, there is a need for an intelligent healthcare assistance system that can:

- Collect symptoms systematically.
- Analyze symptom combinations.
- Predict possible diseases.
- Estimate health risks.
- Provide preliminary recommendations.
- Maintain patient history.
- Generate health reports.
- Present useful healthcare analytics.

MedAssist AI addresses this problem by integrating these capabilities into a single platform.

2.2 Existing System

In a conventional approach, users generally depend on:

- Manual consultation
- Internet searches
- Separate healthcare applications
- Paper-based medical records
- Individual diagnostic services

Such approaches may not provide a unified workflow for symptom analysis, prediction, risk assessment, and reporting.

2.3 Limitations of Existing Approach

1. Fragmented Information - Health information may be distributed across multiple systems.

2. Lack of Automated Analysis - Manual interpretation is required for many preliminary assessments.

3. Limited Personalization - Generic online health information may not consider a user's complete symptom combination or profile.

4. Difficulty in Risk Prioritization - Users may not know which symptoms require urgent professional attention.

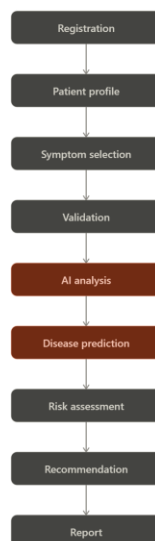
5. Lack of Integrated Reporting - Symptom analysis, predictions, and health summaries may not be available in one centralized report.

2.4 Proposed System

MedAssist AI proposes an integrated AI-assisted healthcare platform.

The proposed system collects user information and symptoms and processes them through multiple modules.

The workflow is:



2.5 Scope of the Project

The scope of MedAssist AI includes:

Patient Management

- Registration
- Login
- Profile management
- Medical history

- Health assessments

Symptom Analysis

- Symptom selection
- Validation
- Correlation analysis
- Historical information integration

Disease Prediction

- Disease classification
- Multiple possible disease predictions
- Probability scores
- Prediction confidence

Risk Assessment

- Severity evaluation
- Risk calculation
- Risk categorization
- Identification of potentially urgent cases

Recommendations

- Healthcare recommendations
- Preventive care suggestions
- Lifestyle advice
- Follow-up guidance

Reporting

- Disease prediction reports
- Risk reports
- Patient summaries
- Downloadable reports

Analytics

- Disease prediction analytics
- Symptom trends

- Healthcare insights
- System monitoring

These modules are explicitly identified in the supplied project specification.

2.6 Project Limitations

MedAssist AI should be considered an assistive and preliminary healthcare system, not a replacement for qualified medical professionals.

The prediction model is dependent on the quality and representativeness of its training data.

The system may not correctly identify conditions that are absent from the training dataset.

Real-world medical diagnosis also requires clinical examination, laboratory tests, imaging, medical history, and professional judgment.

Therefore, predictions generated by the platform should be interpreted as preliminary information.

CHAPTER 3 — SYSTEM ARCHITECTURE

3.1 Introduction

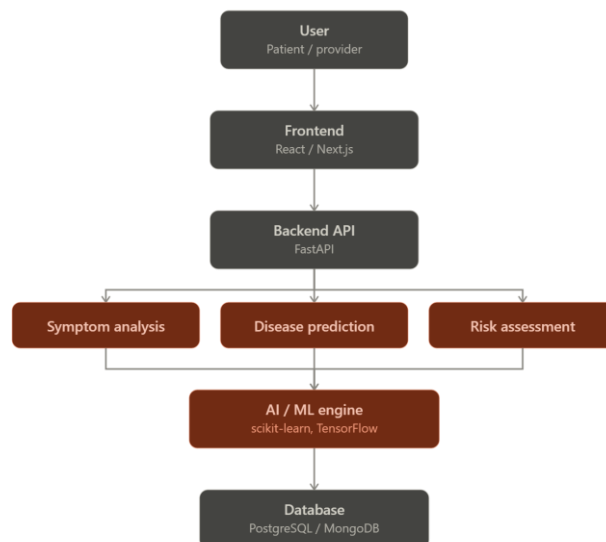
System architecture defines how different components of MedAssist AI interact with one another.

The platform follows a multi-layer architecture consisting of:

1. Presentation Layer
2. Application Layer
3. AI/ML Layer
4. Data Layer
5. Deployment Layer

3.2 High-Level Architecture

You can create your architecture diagram using this structure:



3.3 Presentation Layer

The presentation layer is responsible for user interaction.

The frontend can be developed using:

- React.js
- Next.js
- Tailwind CSS

- Chart.js

The frontend provides:

- Login page
- Registration page
- Patient dashboard
- Symptom interface
- Prediction results
- Risk results
- Reports
- Analytics
- Profile management

The specified technology stack identifies React.js/Next.js and Tailwind CSS for the frontend.

3.4 Application Layer

The application layer contains the backend services.

FastAPI is proposed as the backend framework.

It manages:

- API requests
- Authentication
- Patient information
- Symptom processing
- Model interaction
- Risk assessment
- Recommendations
- Reports

3.5 AI/ML Layer

The AI/ML layer represents the intelligence of the system.

It receives processed symptom information and generates disease prediction results.

Potential technologies include:

- Scikit-learn
- XGBoost
- TensorFlow
- Pandas
- NumPy

These tools are listed in the original project specification.

3.6 Database Layer

The platform can use:

PostgreSQL

Suitable for structured information such as:

- User accounts
- Patient profiles
- Roles
- Assessment records
- Reports

MongoDB - Suitable for flexible or document-oriented healthcare information.

The project specification lists both PostgreSQL and MongoDB as database technologies.

3.7 Security Architecture

The system should implement:

- User authentication
- JWT-based authentication
- Password protection
- Role-based access control
- API authorization
- Controlled access to patient information

JWT authentication is included in the proposed technology stack

CHAPTER 4 — TECHNOLOGY STACK & MODULES

4.1 Technology Stack

Layer	Technology
Frontend	React.js / Next.js
UI	Tailwind CSS
Backend	Python / FastAPI
Database	PostgreSQL / MongoDB
ML	Scikit-learn
Advanced ML	XGBoost
Deep Learning	TensorFlow
Data Processing	Pandas, NumPy
Authentication	JWT
Charts	Chart.js
Containerization	Docker
Cloud	AWS / Azure
API Testing	Postman
IDE	VS Code
Version Control	Git + GitHub

This stack follows the technologies specified in the supplied project document.

4.2 User Management Module

The User Management Module controls user access.

Functions

- Registration
- Login

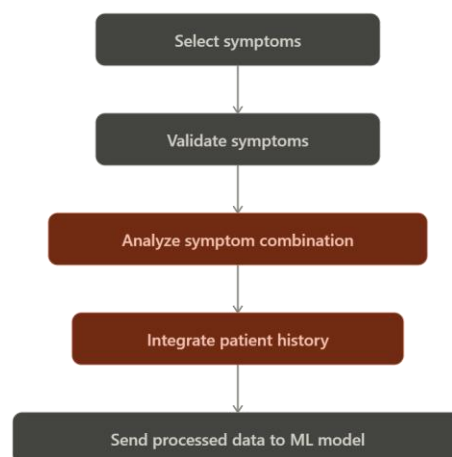
- Logout
- Profile management
- Role management
- Medical history
- Authentication

The system can distinguish between different user roles and provide appropriate dashboard access.

4.3 Symptom Analysis Module

The Symptom Analysis Module is responsible for collecting and processing symptoms.

Workflow:



The project specification includes symptom selection, validation, symptom correlation analysis, and medical history integration.

4.4 Disease Prediction Module

The Disease Prediction Module is the central AI component.

Its major functions include:

- Disease classification
- Probability generation
- Multiple disease prediction
- Confidence scoring

The output can be displayed as:

Disease	Prediction Score
Disease A	82%
Disease B	11%
Disease C	7%

Note: These numbers are illustrative UI examples and should not be presented as actual model results unless obtained during testing.

4.5 Risk Assessment Module

This module evaluates the possible health risk.

Functions include:

- Severity evaluation
- Risk calculation
- Emergency case identification
- Risk categorization

Possible categories can be represented as:

Low Risk → Moderate Risk → High Risk

The exact thresholds should be defined during implementation and validated against the chosen project methodology.

4.6 Treatment Recommendation Module

The recommendation module provides preliminary guidance.

It may include:

- General healthcare recommendations
- Preventive care
- Lifestyle suggestions
- Follow-up guidance

Recommendations should be clearly presented as informational guidance rather than definitive medical treatment.

4.7 Health Reports Module

The report module converts analysis results into structured health summaries.

A report can contain:

1. Patient information
2. Selected symptoms
3. Prediction results
4. Confidence/probability
5. Risk category
6. Recommendations
7. Date and assessment information

4.8 Analytics Dashboard

The analytics module provides visual information such as:

- Disease prediction statistics
- Symptom trends
- Healthcare insights
- System performance

Chart.js can be used to visualize the analytics.

CHAPTER 5 — DATASET & AI/ML MODEL

5.1 Introduction

Machine learning requires relevant and structured data to identify patterns and generate predictions.

The project specification recommends healthcare datasets for three major purposes:

- Symptom analysis
- Disease prediction
- Risk assessment

5.2 Disease Symptoms and Patient Profile Dataset

The recommended primary dataset is the Disease Symptoms and Patient Profile Dataset.

Its purpose includes:

- Symptom-to-disease prediction
- Disease classification
- Medical recommendation workflows

Example disease categories listed in the source include:

- Diabetes
- Malaria
- Dengue
- Hypertension
- Asthma

The document recommends this dataset as the primary dataset because it is relatively accessible and suitable for classification-oriented academic projects.

5.3 Disease Prediction Using Symptoms Dataset

The second proposed dataset supports multi-class disease prediction.

It contains:

- 100+ symptoms
- Disease categories

- Symptom combinations

Its primary applications are:

- Disease Prediction
- Symptom correlation
- Risk assessment

These characteristics are specified in the source document.

5.4 CDC BRFSS Dataset

The CDC Behavioral Risk Factor Surveillance System (BRFSS) dataset can support health risk assessment and population-level analytics.

The source identifies:

- Lifestyle indicators
- Health conditions
- Demographic information

as relevant information within this dataset.

5.5 MIMIC-IV Dataset

MIMIC-IV is identified as an optional advanced dataset.

It can support:

- Clinical decision support
- Healthcare analytics
- Patient risk prediction

The source notes that it contains electronic health records, diagnoses, observations, and clinical measurements. It also notes that credentialed access is required and that it may not be necessary for a student-level implementation.

5.6 Dataset-to-Module Mapping

Dataset	Symptom Analysis	Disease Prediction	Risk Assessment	Analytics
Disease Symptoms Dataset	Yes	Yes	Yes	No

Dataset	Symptom Analysis	Disease Prediction	Risk Assessment	Analytics
Disease Prediction Dataset	Yes	Yes	Yes	No
CDC BRFSS	No	No	Yes	Yes
MIMIC-IV	Yes	Yes	Yes	Yes

This mapping follows the project specification.

5.7 Data Preprocessing

Before training a machine learning model, healthcare data must be prepared.

Typical preprocessing stages include:

1. Data Cleaning

Remove invalid, duplicated, or inconsistent records.

2. Missing Value Handling

Identify missing values and apply an appropriate strategy.

3. Feature Selection

Select relevant symptoms and patient characteristics.

4. Encoding

Convert categorical information into machine-readable numerical representations.

5. Data Transformation

Transform the dataset into the format required by the ML algorithm.

6. Train-Test Split

Divide the dataset into training and testing portions.

5.8 Feature Representation

Symptoms can be represented using binary encoding.

For example:

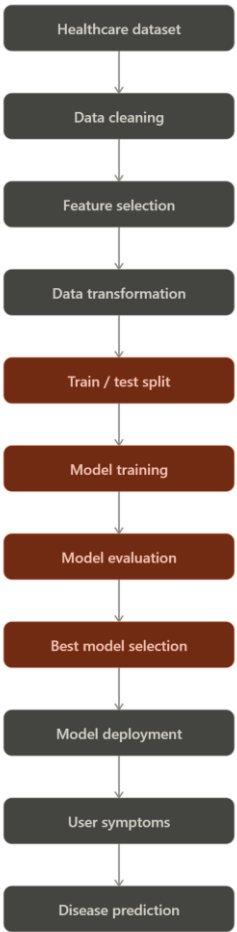
Symptom	Value
Fever	1
Headache	1
Cough	0
Fatigue	1
Nausea	0

Here:

- **1** = symptom present
- **0** = symptom absent

This representation allows classification algorithms to process symptom information.

5.9 Machine Learning Pipeline



5.10 Candidate ML Algorithms

Potential classification algorithms include:

Decision Tree

A decision tree makes predictions using a sequence of decision rules.

Random Forest

Random Forest combines multiple decision trees to improve classification performance.

XGBoost

XGBoost is a gradient boosting framework suitable for structured classification problems.

Neural Networks

TensorFlow can be used to implement neural network models if sufficient data and computational resources are available.

The project specification includes Scikit-learn, XGBoost, and TensorFlow as AI/ML technologies.

5.11 Model Training

The model-training process includes:

1. Loading the dataset.
2. Cleaning the data.
3. Selecting features.
4. Encoding categorical information.
5. Splitting data into training and testing sets.
6. Training candidate models.
7. Evaluating models.
8. Selecting the appropriate model.
9. Saving the trained model.
10. Integrating it with the FastAPI backend.

5.12 Model Evaluation Metrics

The project defines the following AI performance metrics:

- Accuracy
- Precision
- Recall
- F1-score

The project also identifies healthcare-specific metrics such as disease prediction confidence, risk assessment accuracy, and recommendation relevance.

CHAPTER 6 — IMPLEMENTATION & MILESTONES

6.1 Development Methodology

The project can be implemented incrementally over eight weeks.

The source document divides implementation into four major milestones.

6.2 Milestone 1 — Week 1 & 2

Project Initialization

Activities:

- Define objectives
- Define healthcare workflows
- Design system architecture
- Design database schema
- Create UI wireframes
- Set up frontend
- Set up backend
- Implement authentication
- Integrate datasets
- Create symptom management workflow

Expected outcomes include project architecture, authentication, patient dashboard, dataset integration, and symptom management.

6.3 Milestone 2 — Week 3 & 4

This phase focuses on AI functionality.

Activities:

- Prepare training data
- Train disease prediction models
- Evaluate models
- Integrate the selected model
- Generate disease probability results

- Build risk scoring engine
- Implement severity analysis
- Generate health risk reports

The source specifies disease prediction, risk assessment, probability scoring, and severity analysis for this milestone.

6.4 Milestone 3 — Week 5 & 6

This phase focuses on recommendations and analytics.

Activities:

- Develop healthcare recommendations
- Develop preventive-care suggestions
- Implement advisory features
- Create analytics dashboard
- Create disease prediction reports
- Implement trend visualization

The expected outcome is a complete healthcare intelligence workflow connecting prediction, recommendations, reporting, and analytics.

6.5 Milestone 4 — Week 7 & 8

Final implementation focuses on:

- Testing
- Validation
- Performance optimization
- Docker deployment
- Cloud deployment
- Documentation
- Presentation
- End-to-end demonstration

The source specifically identifies Docker and cloud environments for deployment during this stage.

CHAPTER 7 — USER ROLES & DASHBOARD

7.1 User Roles

The platform can use role-based access control.

Possible roles include:

Patient/User

Can:

- Register
- Login
- Manage profile
- Enter symptoms
- View predictions
- View risk assessment
- View recommendations
- View reports
- Review history

Healthcare Provider

Can:

- View authorized patient information
- Review assessment reports
- Examine prediction history
- Analyze health trends
- Support patient follow-up

Administrator

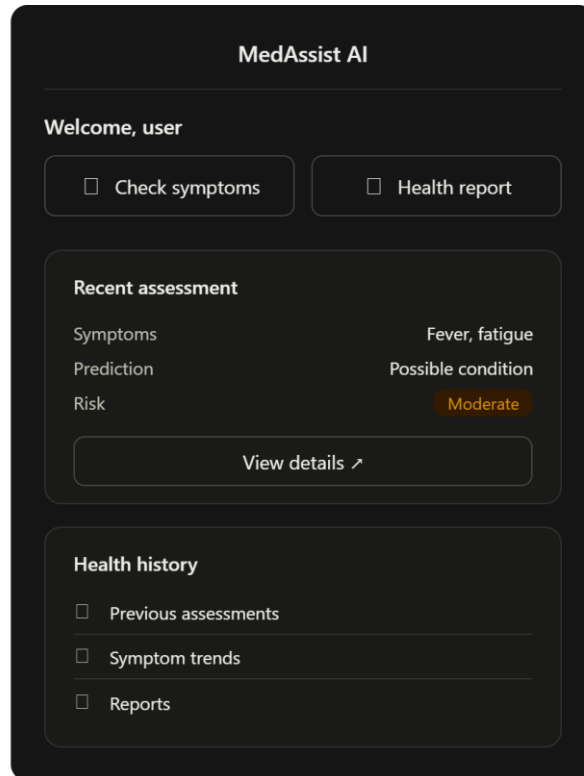
Can:

- Manage users
- Monitor system activity
- Manage platform configuration
- Monitor performance
- View analytics

7.2 Patient Dashboard

The patient dashboard should provide a simple overview.

Possible components:



7.3 Symptom Assessment Screen

The symptom assessment page should allow users to:

1. Search symptoms.
2. Select symptoms.
3. Review selected symptoms.
4. Submit the assessment.
5. Receive prediction results.

7.4 Prediction Dashboard

The prediction dashboard should display:

- Possible diseases
- Probability scores

- Confidence
- Important symptoms
- Risk category

7.5 Risk Dashboard

The risk dashboard can represent:

Risk Level: LOW / MODERATE / HIGH

It can also display:

- Severity assessment
- Factors contributing to risk
- Recommended next steps
- Follow-up guidance

7.6 Healthcare Analytics Dashboard

Healthcare providers or administrators can view:

- Disease prediction statistics
- Symptom trends
- Risk categories
- System performance
- Historical patterns

CHAPTER 8 — TESTING & PERFORMANCE EVALUATION

8.1 Introduction

Testing ensures that the platform functions correctly and reliably.

Testing should cover:

- Frontend / Backend
- Database
- APIs
- ML model
- Authentication
- Reports / Dashboards
- Deployment

8.2 Unit Testing

Individual components should be tested independently.

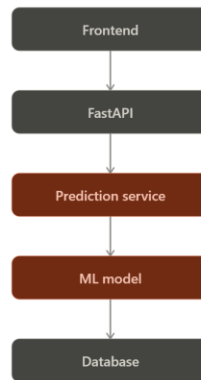
Examples:

- Login validation
- Symptom selection
- Risk calculation
- Prediction function
- Report generation

8.3 Integration Testing

Integration testing verifies whether different modules communicate correctly.

Example:



The complete workflow should produce the expected output.

8.4 API Testing

Postman can be used for API testing.

Example APIs:

API	Purpose
POST /register	Register user
POST /login	Authenticate user
POST /symptoms	Submit symptoms
POST /predict	Generate prediction
POST /risk	Calculate risk
GET /history	Retrieve history
GET /report	Generate report

8.5 Machine Learning Testing

The ML model should be evaluated using:

Accuracy

Measures the proportion of correctly classified instances.

Precision

Measures how many predicted positive cases are actually positive.

Recall

Measures how many actual positive cases are correctly identified.

F1-score

Provides a balance between precision and recall.

8.6 Healthcare Performance

The source identifies:

- Disease prediction confidence
- Risk assessment accuracy
- Recommendation relevance

as important healthcare performance indicators.

8.7 System Performance

System-level performance can be measured using:

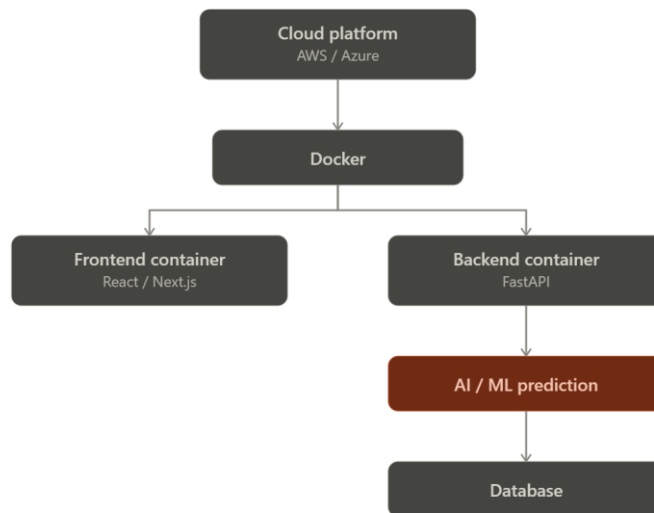
- API response time
- Dashboard loading speed
- Concurrent user handling

These are explicitly identified as platform performance metrics in the project specification.

CHAPTER 9 — DEPLOYMENT & RESULTS

9.1 Deployment Architecture

The application can be containerized using Docker.



Docker and AWS/Azure are included in the proposed deployment stack.

9.2 Docker

Docker packages the application and its dependencies into containers.

Benefits include:

- Consistent environment
- Easier deployment
- Dependency isolation
- Portability
- Simplified scaling

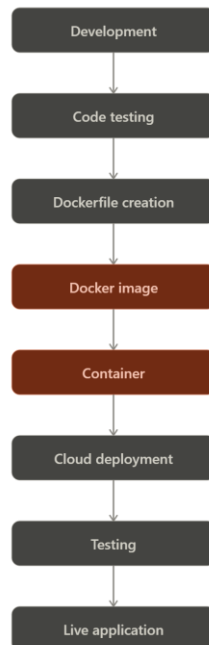
9.3 Cloud Deployment

The platform can be deployed using:

- AWS
- Azure

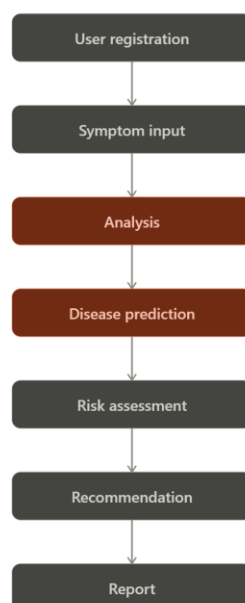
Cloud deployment makes the platform accessible through the internet and provides infrastructure for hosting the frontend, backend, database, and AI services.

9.4 Deployment Process



9.5 Results

The project is expected to demonstrate an end-to-end workflow:



The project specification defines successful completion as a deployed frontend and backend, validated AI models, prepared documentation, and a successful end-to-end demonstration.

9.6 Quantitative Evaluation

The project defines quantitative goals around:

Symptom Analysis

Accurately analyze symptoms and identify potential health conditions.

Disease Prediction

Generate disease predictions with confidence scoring.

Risk Assessment

Assess health risks and identify high-priority cases.

Platform Performance

Support multiple concurrent healthcare assessments while maintaining stable performance.

Important: In your final report, replace these qualitative goals with your actual measured accuracy, precision, recall, F1-score, response time, and other results once your implementation is tested.

CHAPTER 10 — CONCLUSION & FUTURE ENHANCEMENTS

10.1 Conclusion

MedAssist AI presents an integrated approach to AI-assisted healthcare support by combining symptom analysis, disease prediction, risk assessment, healthcare recommendations, health reports, patient history, and analytics into a centralized platform. The system demonstrates how artificial intelligence and modern web technologies can be combined to provide preliminary healthcare insights.

The project also provides practical exposure to:

- Artificial intelligence
- Machine learning
- Healthcare datasets
- REST APIs
- React/Next.js
- FastAPI
- Database management
- Authentication
- Data visualization
- Docker
- Cloud deployment

The proposed system is intended to support informed healthcare decisions rather than replace professional medical diagnosis.

10.2 Future Enhancements

1. Advanced Machine Learning Models - Future versions can evaluate more advanced algorithms and deep learning approaches.

2. Larger Healthcare Datasets - Additional datasets can improve the diversity of conditions represented by the model.

3. Electronic Health Record Integration - Integration with electronic health records could provide more comprehensive patient information.

4. Multilingual Support - The platform could support multiple regional and international languages.

- 5. Voice-Based Symptom Input** - Users could describe symptoms through speech.
- 6. Wearable Integration** - Future versions could integrate data from smart watches and healthcare sensors.
- 7. Real-Time Health Monitoring** - Continuous monitoring could be used for long-term health tracking.
- 8. Explainable AI** - The system could explain which symptoms contributed most strongly to a prediction.
- 9. Advanced Analytics** - Healthcare organizations could use predictive analytics to identify trends and patterns.
- 10. Telemedicine Integration** - The platform could connect users with healthcare professionals after preliminary assessment.
- 11. Improved Risk Prediction** - Future models could incorporate more patient information and longitudinal health history.
- 12. Mobile Application** - A dedicated Android/iOS application could improve accessibility.

CHAPTER 11 – APPENDIX

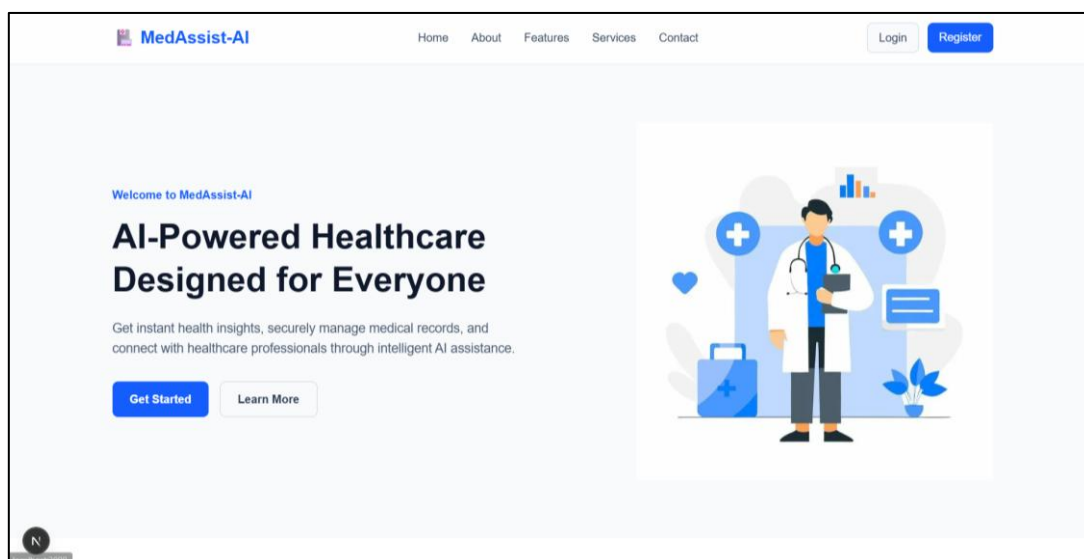
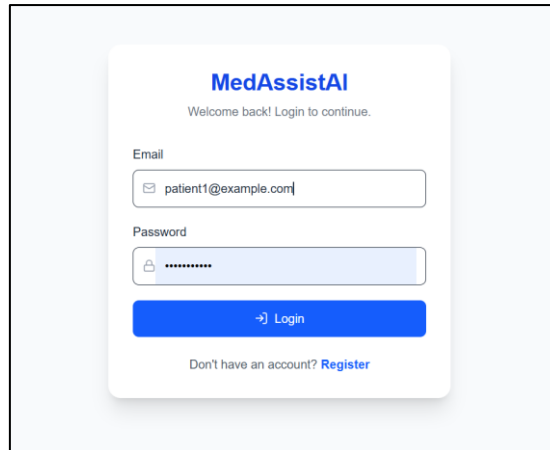
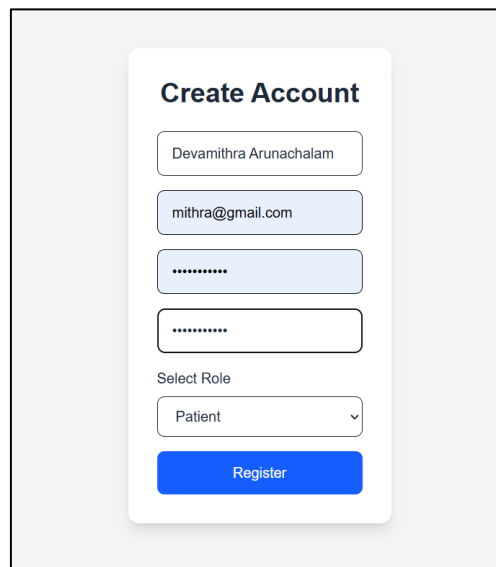


Figure 1: Landing Page – Displays the MedAssist AI homepage introducing the platform and its primary healthcare assistance features.



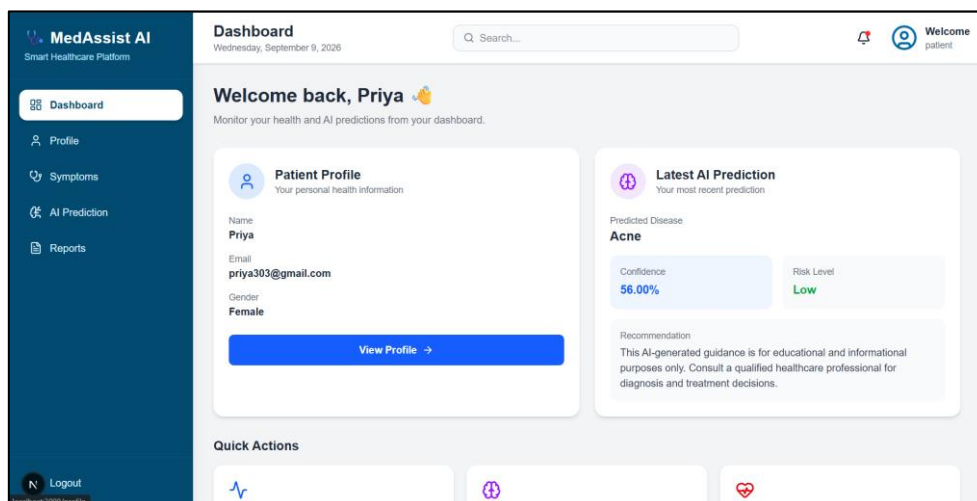
The login page for MedAssistAI features a white card on a light blue background. At the top, the logo 'MedAssistAI' is displayed in blue, followed by the text 'Welcome back! Login to continue.' Below this, there are two input fields: 'Email' with the placeholder 'patient1@example.com' and 'Password' with a masked password '*****'. A blue 'Login' button with a right-pointing arrow is positioned below the password field. At the bottom of the card, a link 'Don't have an account? Register' is provided in blue text.

Figure 2: Login page of MedAssist AI for secure user authentication and system access.



The registration page for MedAssistAI shows a white card on a light blue background. The title 'Create Account' is at the top. Below it are four input fields: a full name field containing 'Devamithra Arunachalam', an email field containing 'mithra@gmail.com', and two masked password fields. A 'Select Role' dropdown menu is set to 'Patient'. A blue 'Register' button is at the bottom of the card.

Figure 3: Registration Page – Displays the user registration form for creating a new MedAssist AI account.



The patient dashboard for MedAssistAI has a dark blue sidebar on the left with navigation links: 'Dashboard' (selected), 'Profile', 'Symptoms', 'AI Prediction', and 'Reports'. The main content area is titled 'Dashboard' and 'Welcome back, Priya'. It features two primary sections: 'Patient Profile' showing personal information (Name: Priya, Email: priya303@gmail.com, Gender: Female) and 'Latest AI Prediction' showing a predicted disease of 'Acne' with a 56.00% confidence and low risk level. A 'Quick Actions' section at the bottom contains icons for a heart rate monitor, a brain, and a heart.

Figure 4: Patient Dashboard – Presents the patient’s personalized dashboard containing health assessment and system navigation options.

Priya
priya303@gmail.com
PATIENT [Edit Profile](#)

Personal Information
Basic information about the patient.

Full Name: Priya
Email Address: priya303@gmail.com
Email cannot be changed here.
Date of Birth: 21-Dec-2006
Gender: Female

Contact Information
Contact and emergency information.

Phone Number: 9876543432
Emergency Contact: 9898976565

Figure 5: Profile Page – Displays the patient profile containing personal information and relevant health details.

Patient Symptoms
Record and manage patient symptoms.

Add Symptoms

Fever Cough
Headache Fatigue
Chest Pain Shortness of Breath
Blood Pressure Heart Rate
Blood Sugar Temperature
Notes
[Save Symptoms](#)

Figure 6: Symptom Selection – Shows the interface where users select and submit their current symptoms for analysis.

Disease Prediction
Create, update and manage patient disease predictions.

AI Disease Prediction
Select your symptoms and let MedAssist AI predict possible diseases.

Search Symptom
[Search symptoms...]
Selected Symptoms (3)
High Fever Cold Hands And Feet Cough
[Predict Disease](#)

Prediction History
View and manage your previous AI predictions. [2 Predictions](#)

Disease	Confidence	Risk Level	Actions
Acne Blackheads	56%	Low	View Report Edit Delete

Figure 7: Symptom Analysis – Displays the system’s analysis of the selected symptoms and their possible health-condition associations.

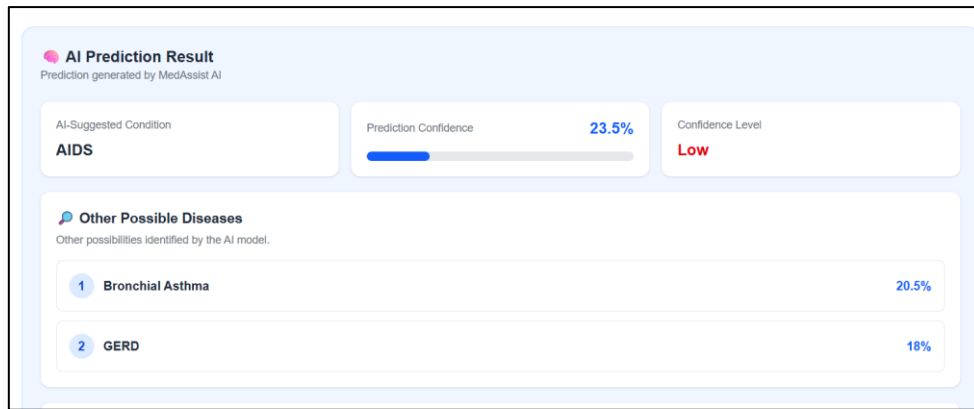


Figure 8: Disease Prediction – Shows the AI-generated possible disease predictions along with corresponding confidence or probability scores.



Figure 9: Risk Assessment – Displays the assessed health-risk category and severity information based on the user's symptoms and profile.

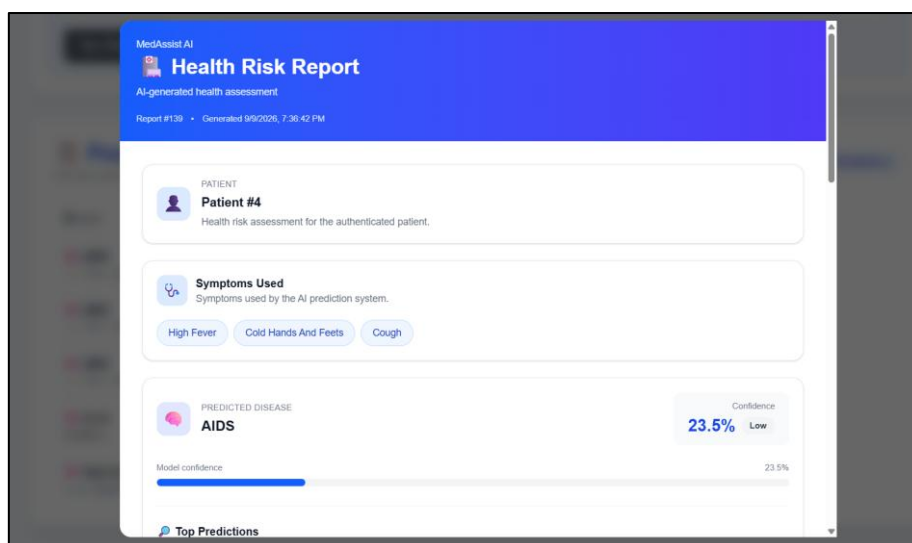


Figure 10: Health Report – Displays the generated health report containing symptom analysis, prediction, risk assessment, and recommendations.

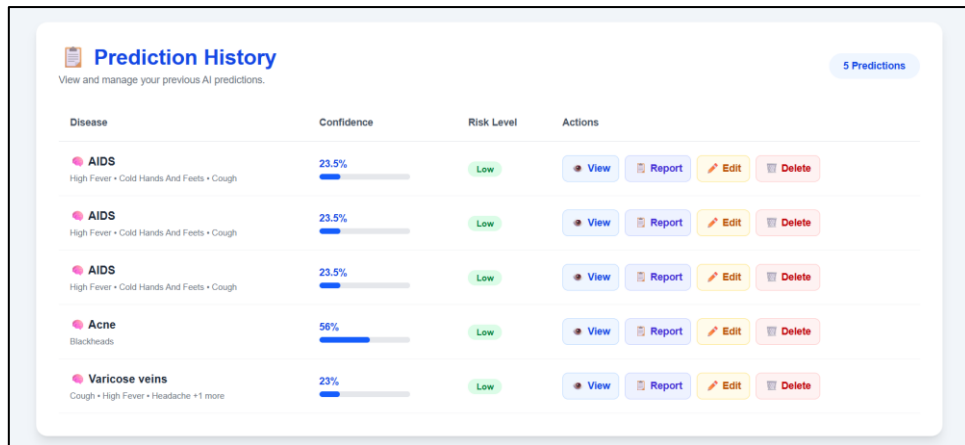


Figure 11: Patient History – Shows the user’s previous health assessments and historical prediction records.

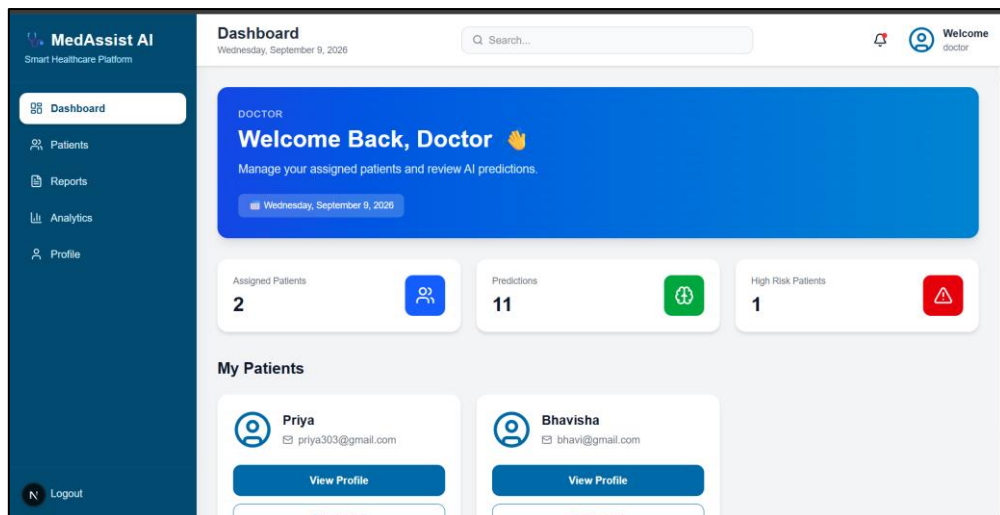


Figure 12: Doctor Dashboard – Shows the number of patients assigned for a doctor in doctor dashboard

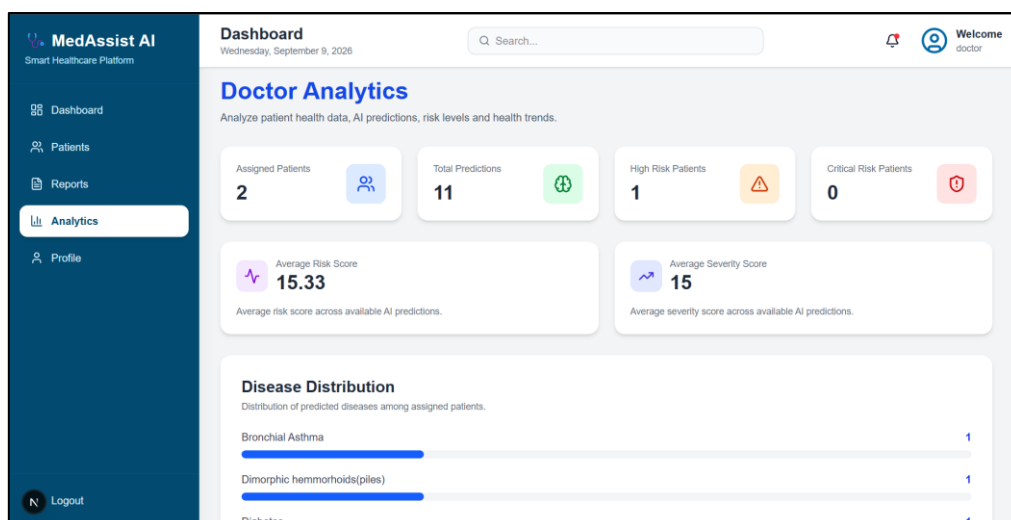


Figure 13.a: Analytics Dashboard

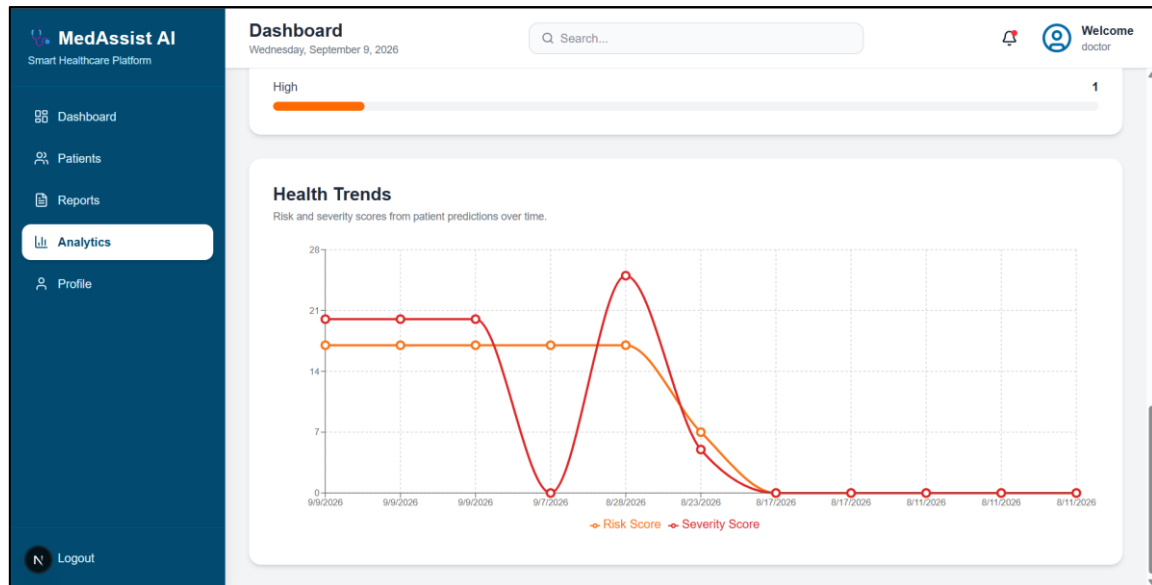


Figure 13.b: Analytics Dashboard – Presents graphical analytics related to disease predictions, symptom trends, and healthcare insights.

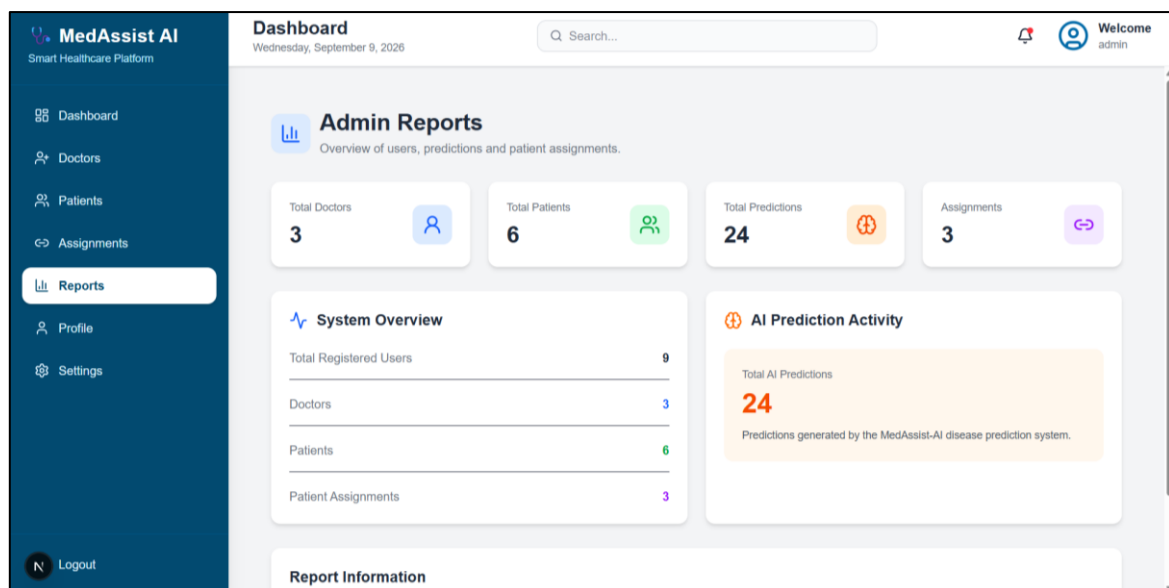


Figure 14: Admin Dashboard – Displays administrative controls and system-level analytics for monitoring the MedAssist AI platform.

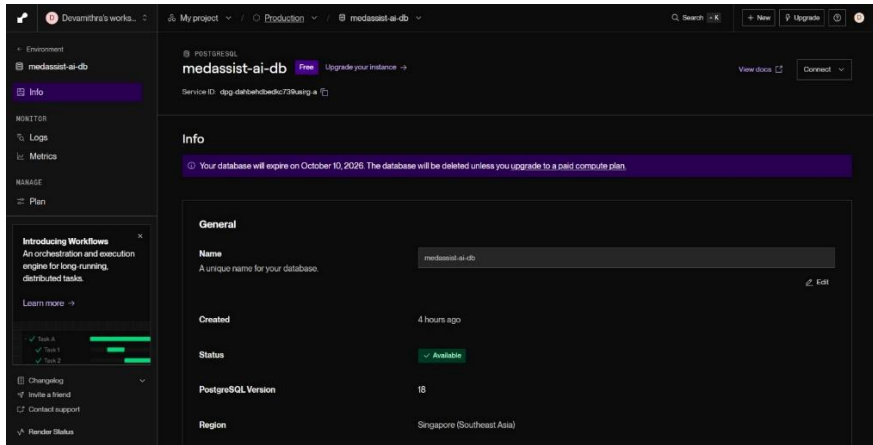


Figure 15.a: Render Database

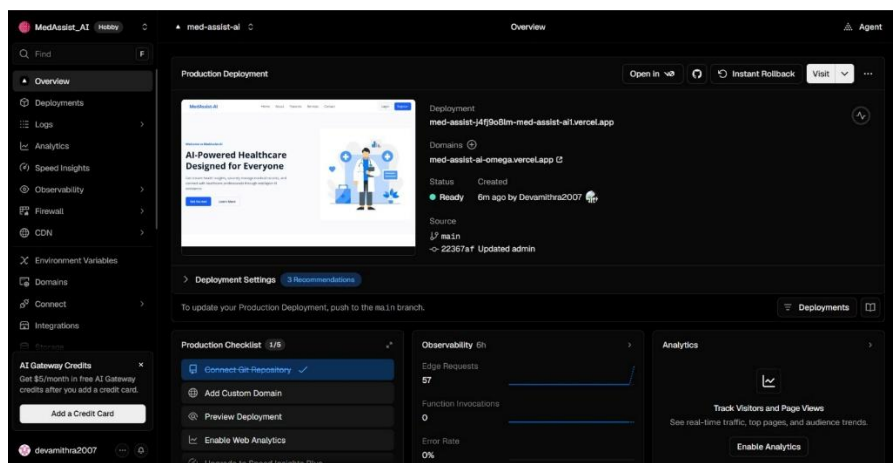


Figure 15.b: Vercel Frontend Deployment

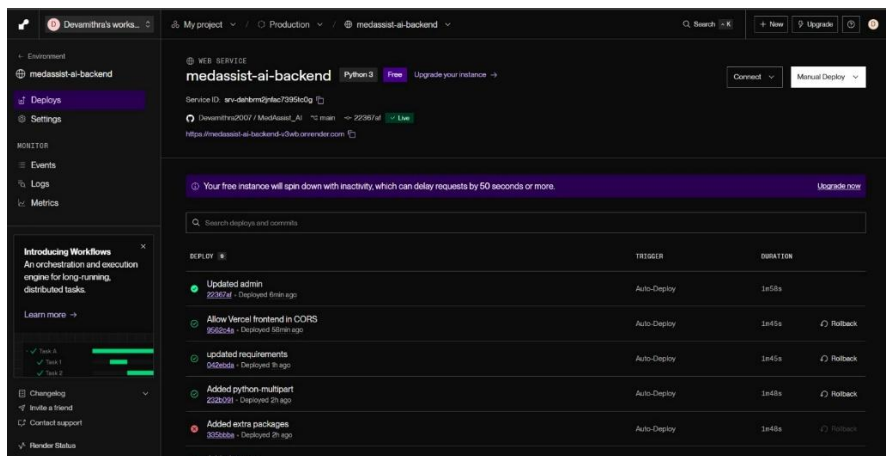


Figure 15.c: Render Backend Deployment – Displays the MedAssist AI application containers running successfully through Docker.